

Statistics for Human Genetics

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What can we do with Statistics ?

1. Introduction
2. Hypothesis testing
3. GWAS results and in practice
4. Prediction: PRS

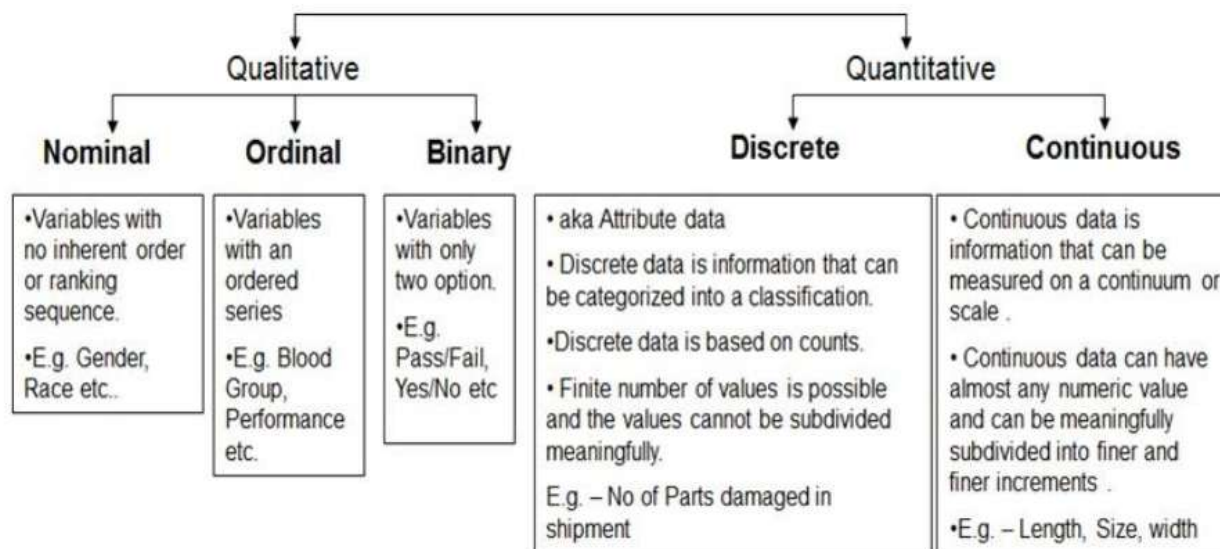
1

Random variables and estimation



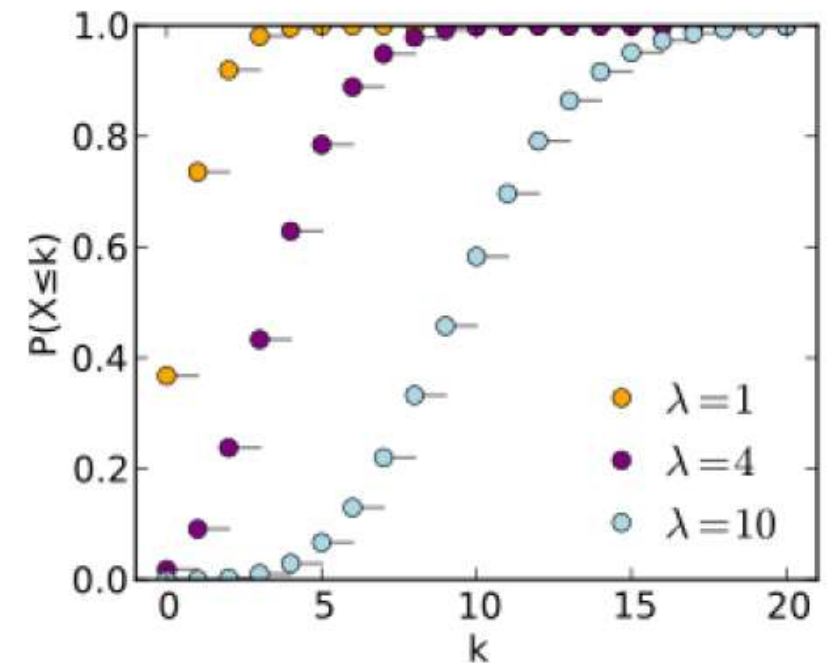
Random Variables

- In statistics, we measure the realizations/observations of random variables
- Often, these random variables follow a distribution
- They can be qualitative or quantitative (continuous or discrete)



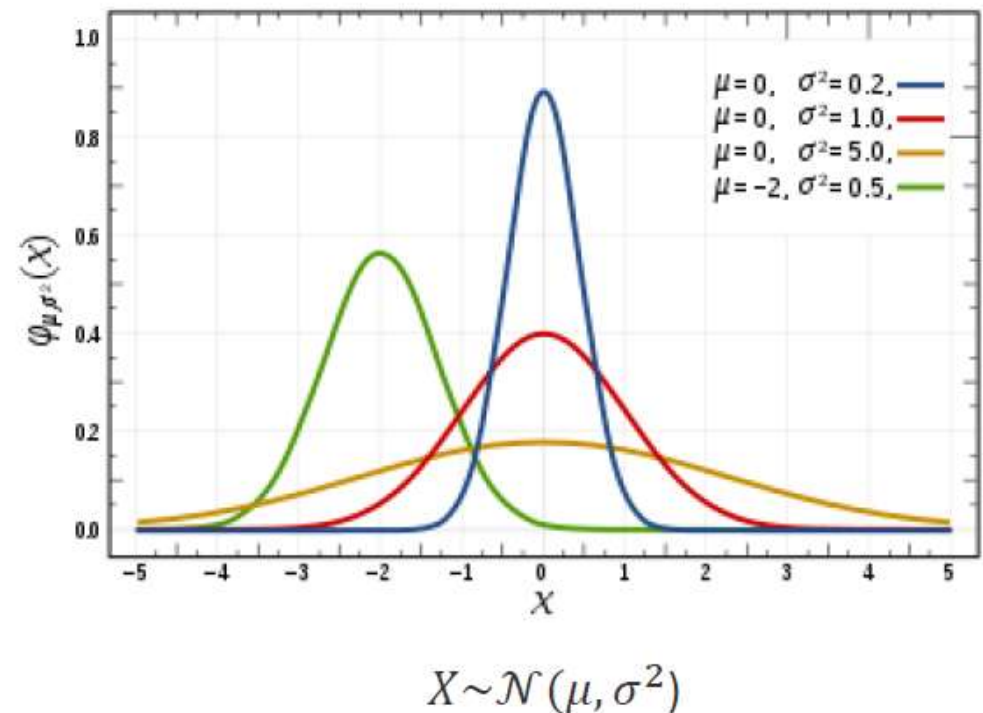
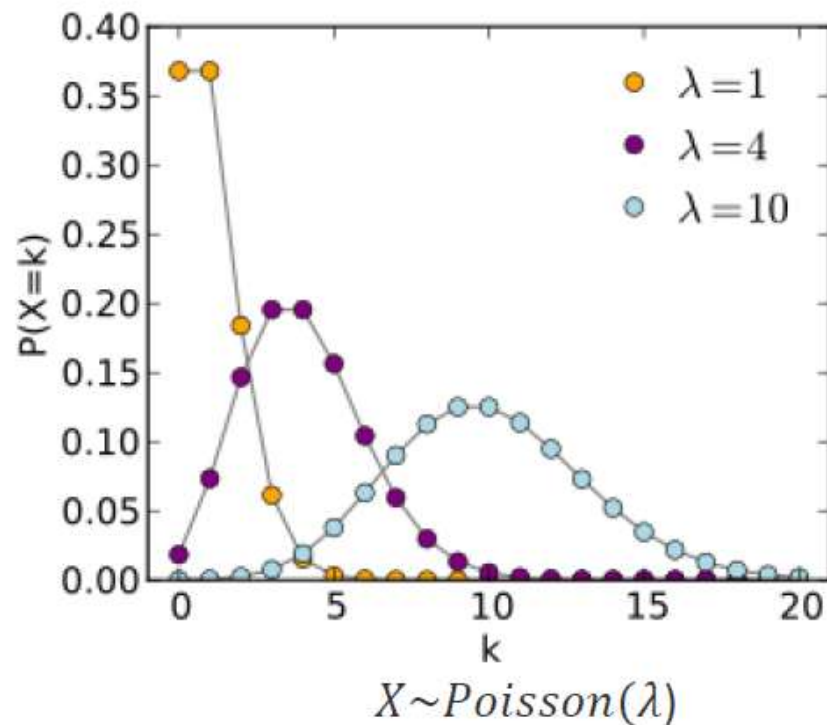
Representations – Cumulative distribution function (CDF)

- $y = p(X \leq x)$
- Always growing
- Ideal way to represent but hard to read
- All distributions look the same

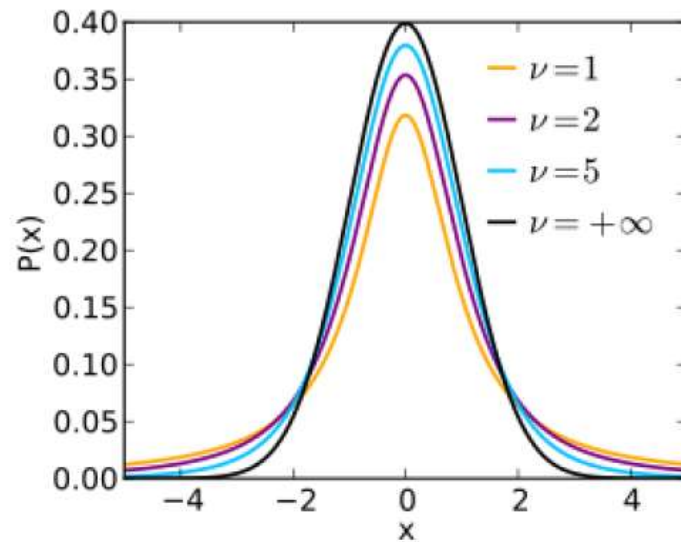


Representations – Probability density function

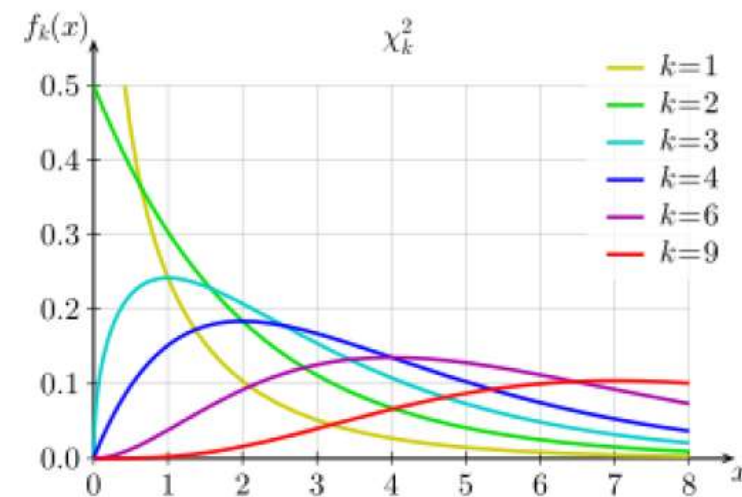
- $y = p(X = x)$ for discrete
- Shows how values are distributed
- Nice visually but hard to deal with



Distributions – tests statistics



$$X \sim T(\nu)$$



$$X \sim \chi^2(k)$$

λ , μ , σ , ν and k are the ideal, theoretical parameters
→ Use estimations from the data to approximate them

Statistics

- A statistics is a meaningful quantity derived from the data
- Often, estimators are realizations of distribution parameters
- Examples include mean, proportion, ...
- For simple distributions/parameters, there is a formula
- For more complex ones, we have to use other techniques (Monte-Carlo, Permutations, ...)

$$\hat{p} = \frac{x}{n}$$

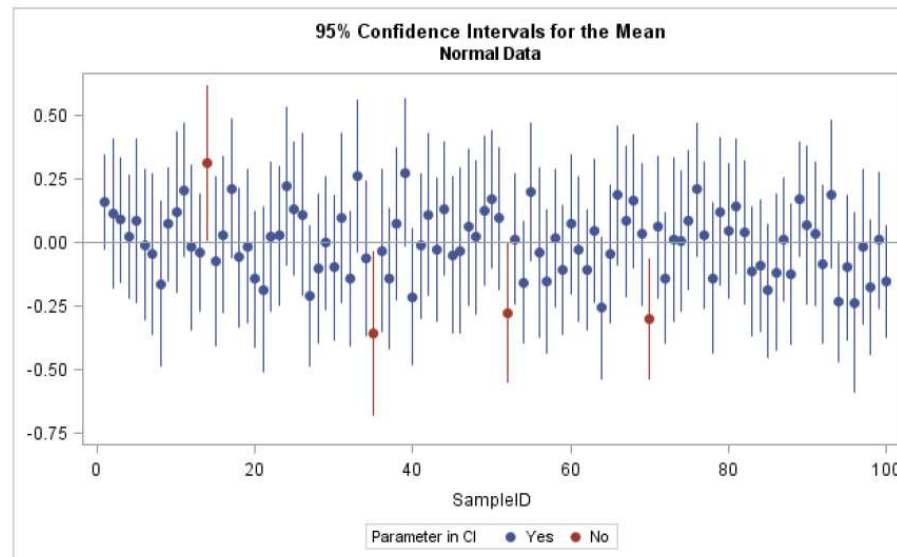
$$(\hat{\mu} =) \bar{x} = \frac{1}{n} \sum_{k=0}^n x_k$$

$$w = \frac{(\hat{\theta} - \theta_0)^2}{se(\hat{\theta})} \sim \mathcal{N}(0,1)$$

$$(\widehat{\sigma^2} =) s^2 = \frac{1}{N-1}$$

Confidence intervals

- $x\%$ confidence interval ($x\%C.I.$):
 - $x\%$ of the time when this interval is calculated, it will contain the true value of the parameter
 - The true value of the parameter has $x\%$ chances to be in the $x\%C.I.$
- Often 95% $C.I.$ used corresponding to the classical α level



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Modelling



Modelling

- Estimate the effect of one variable on another variable

$$\text{phenotype} \sim \beta \times \text{genotype} + \epsilon$$

$$\begin{array}{ccc} \begin{bmatrix} \text{phenotype}_0 \\ \vdots \\ \text{phenotype}_n \end{bmatrix} & \begin{bmatrix} A/T \\ \vdots \\ T/T \end{bmatrix} & \begin{bmatrix} 1 \\ \vdots \\ 2 \end{bmatrix} \\ = \{0,1\} \text{ (case-control)} & = \{0,1,2\} \text{ (genotype, directly typed)} & \\ \in \mathbb{R} \text{ (quantitative)} \sim \mathcal{N}(0,1) & \in [0,2] \text{ (dosage, imputed)} & \begin{bmatrix} 0.965 \\ \vdots \\ 1.816 \end{bmatrix} \end{array}$$

- What is the effect of the genotype on the phenotype ?
 - β estimations for continuous phenotypes
 - Odds Ratio (OR) for binary phenotypes. Ex: Case/Control studies

2

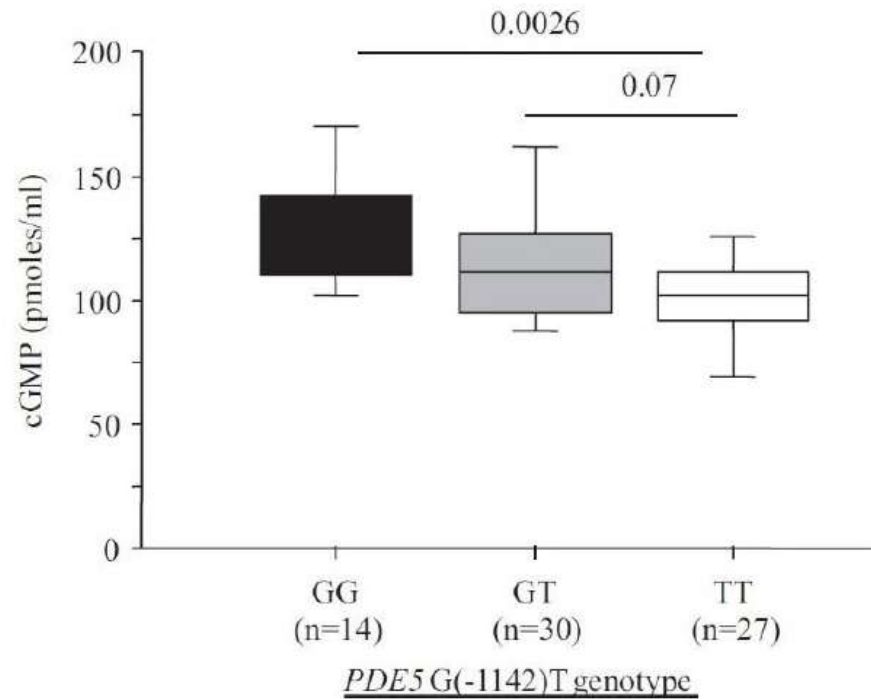
Modelling

2.1 – Continuous traits

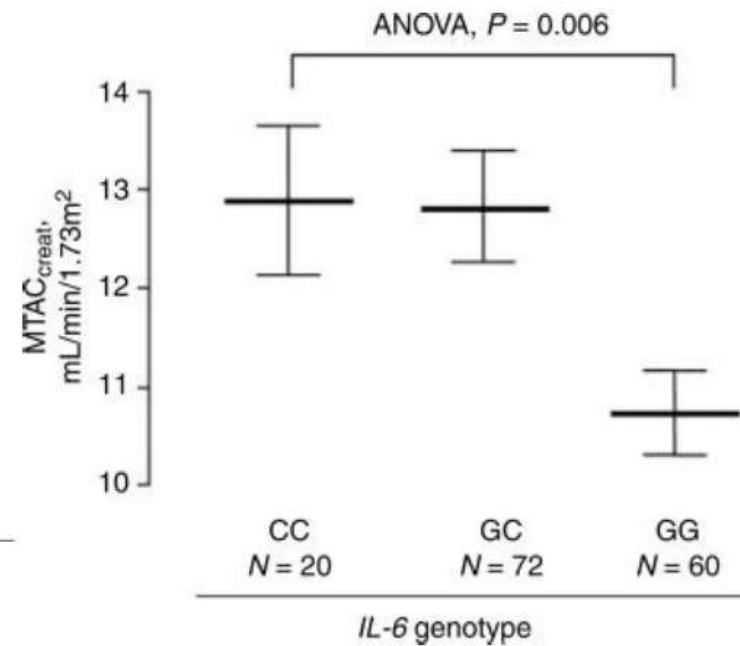


Continuous traits

- If directly typed genotypes (0, 1, 2) are analyzed: ANOVA



Additive

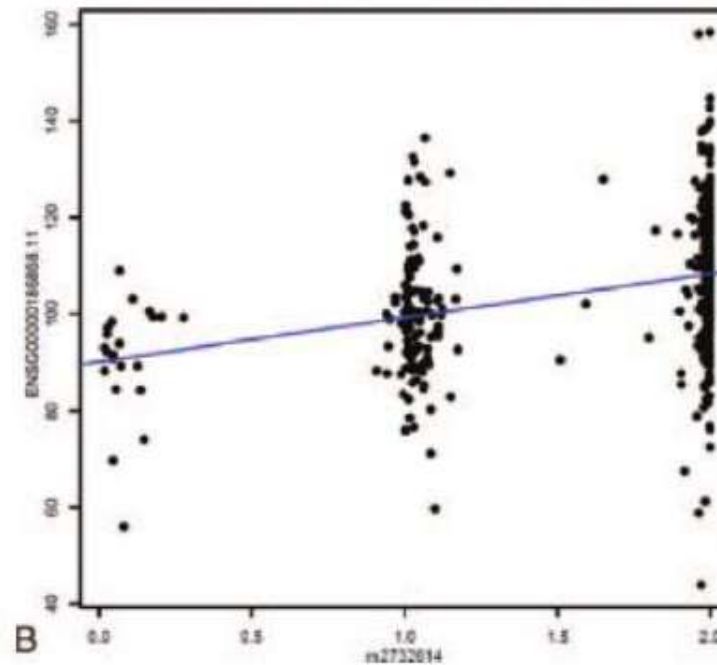
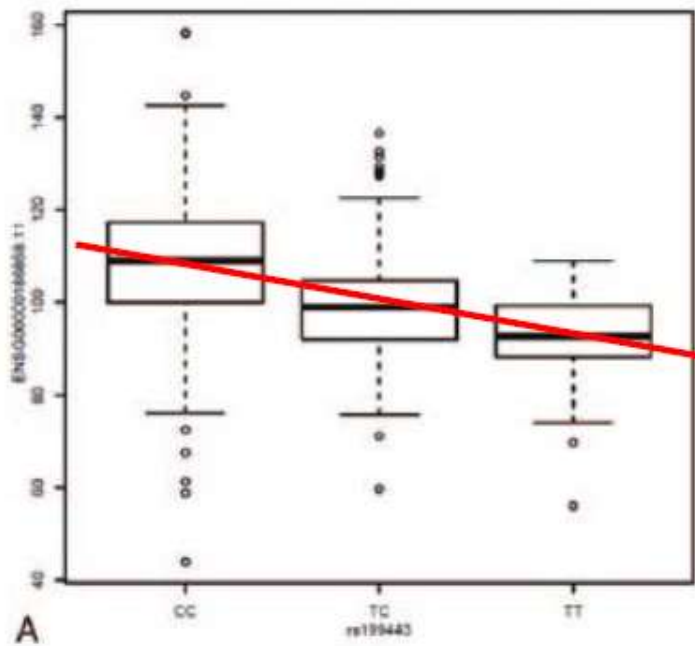


Recessive

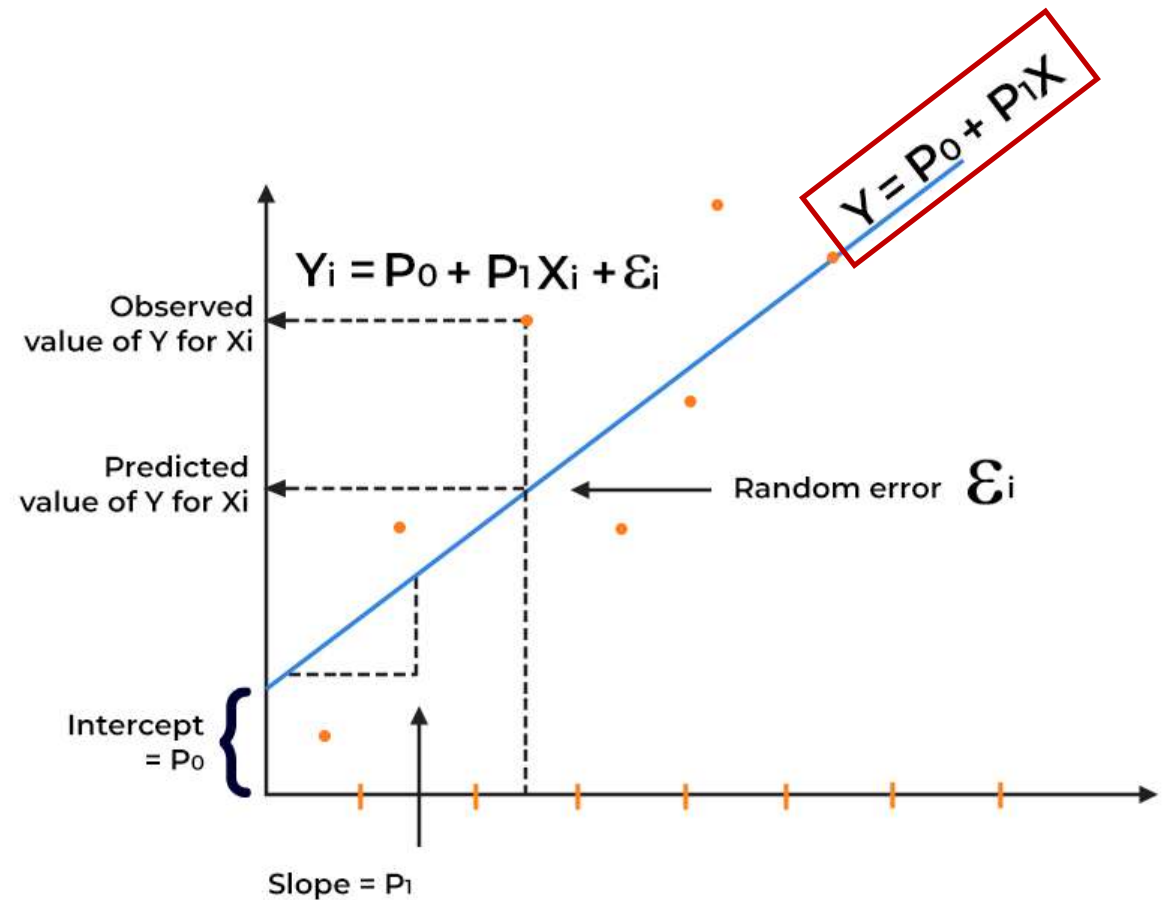
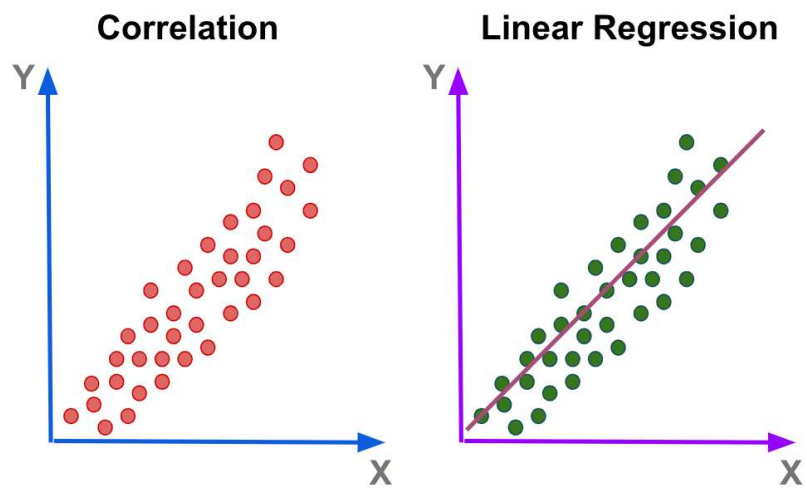
HELMHOLTZ MUNICH

Continuous traits

- If dosages are analyzed (imputed quantity of minor allele $d \in [0,1]$): linear regression
- In general: generalized linear models



Linear regression

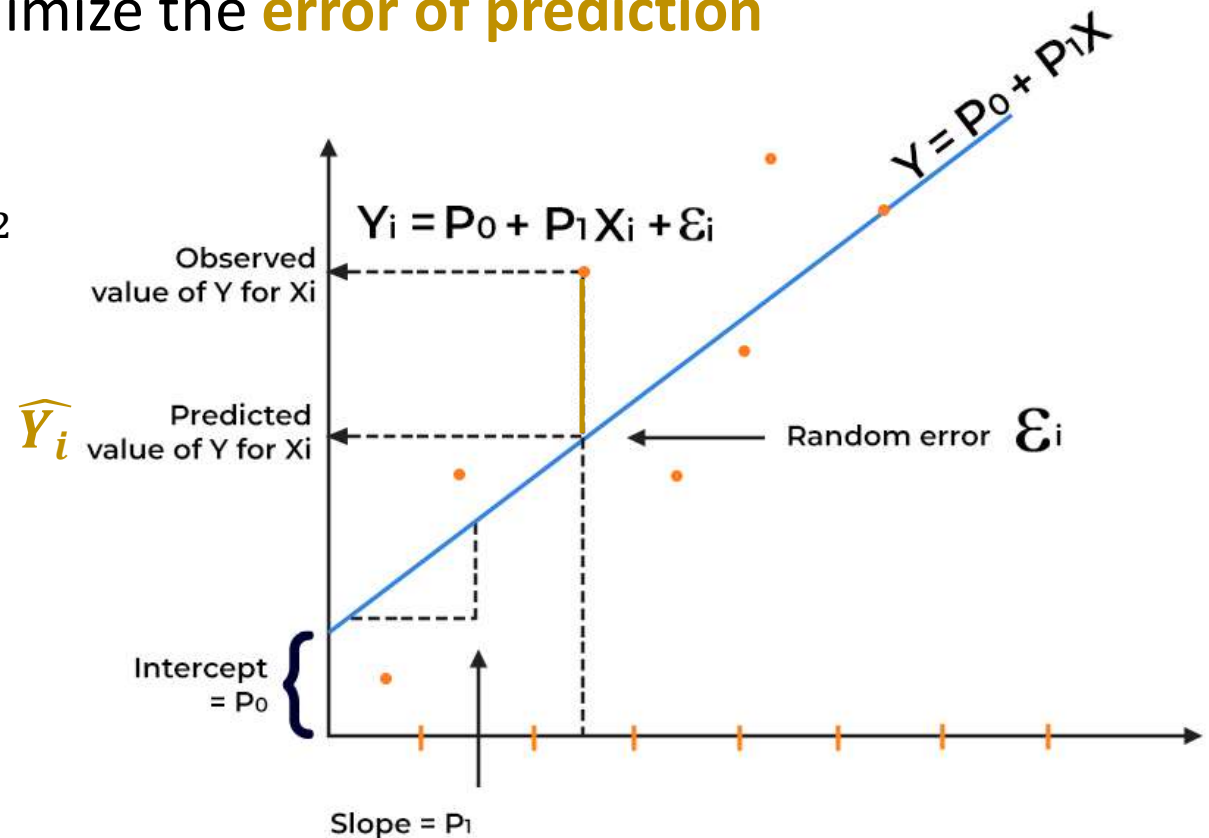


- Straight lines characterized by slope and intercept
- Slope: Change in Y due to change of one unit in X
- Intercept: value of Y when $X = 0$

Linear regression – Least square regression

- Least square regression: minimize the **error of prediction**

- Minimize $\sum (\hat{Y}_i - Y_i)^2$
- Minimize $\sum (P_1 X_i + P_0 - Y_i)^2$



Continuous traits

- A linear regression model is defined as:

$$y = x\beta_1 + \beta_0 + \varepsilon$$

- Data:

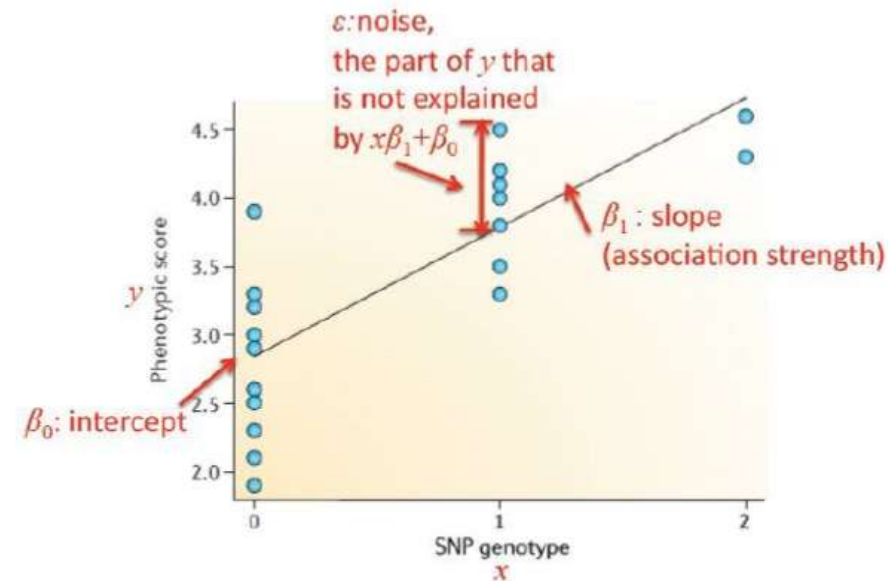
- y is a continuous trait
- x is the SNP genotype at a given locus

- Parameters:

- β_1 is the regression coefficient, represents the strength of association between y and x
 - $\beta_1 > 0$: for every supplementary allele, the phenotype will increase by the beta coefficient value
 - $\beta_1 < 0$: for every supplementary allele, the phenotype will decrease by the beta coefficient value
- β_0 : intercept term (is often ignored)

- Assumptions:

- The individuals in the study are not related
- The phenotype y has a normal distribution



2

Modelling

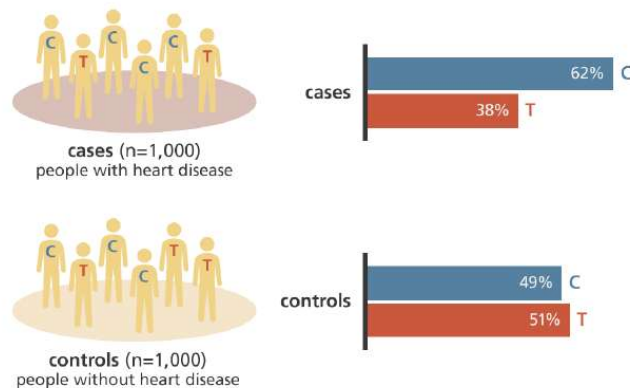
2.2 – Case/Control



Case/control studies

- OR: *how much more likely are you to be a case if you carry the risk allele ?*

➤ Per genotype g and disease Y , we compute the odds $O = \frac{p}{1-p} = \frac{p_{Y=1|g}}{1-p_{Y=1|g}}$



	Cases	Controls	N
T	380	510	890
C	620	490	1100

$$O_T = \frac{380/890}{510/890} \quad O_C = \frac{620/1100}{490/1100}$$

- OR: Ratio of the odds of the two alleles

- OR>1: the allele is 'deleterious'
- OR<1: the allele is 'protective'

$$OR_{C/T} = \frac{620 * 510}{490 * 380} = 1.70$$

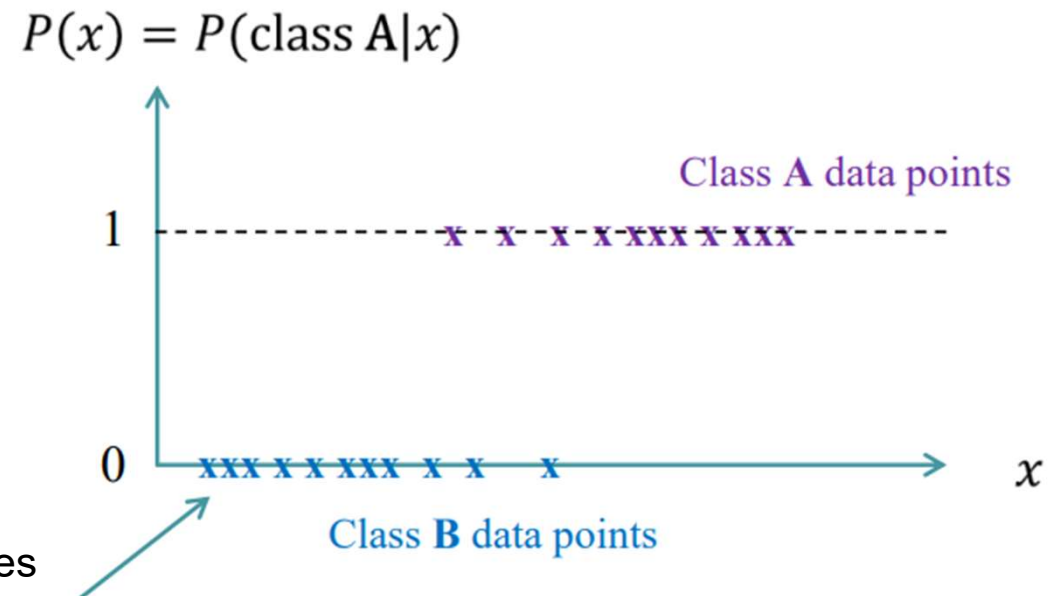
Case/control studies

- Output: OR and 95% confidence interval of the OR
- Association test: is it significantly different from 1 ?
 - $H_0: OR = 1$
 - $H_1: OR \neq 1$
- Statistics: Fisher's exact test or Chi-squared
- In case of dosages or if covariates are included: logistic regression

Binary traits

- Example: disease status with cases and controls
- What would be a good model?

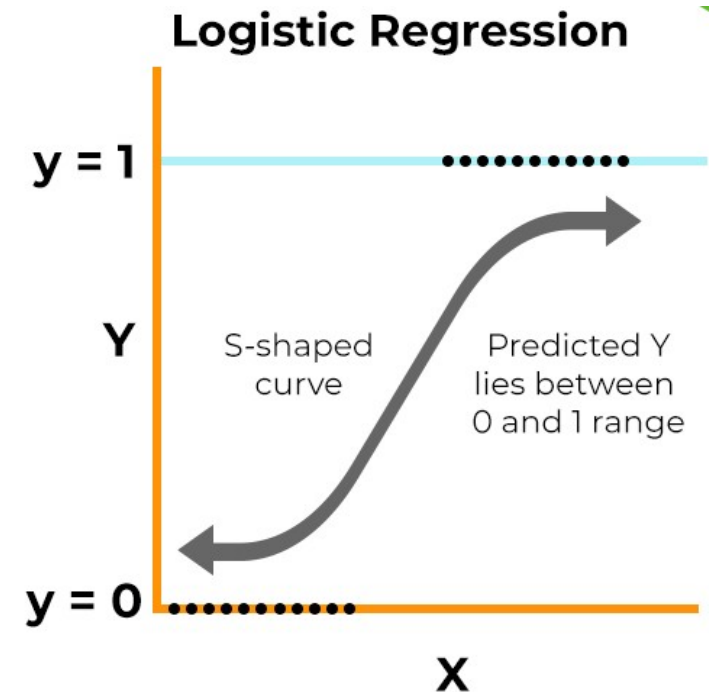
These are **not** probabilities, just the x values of the data points



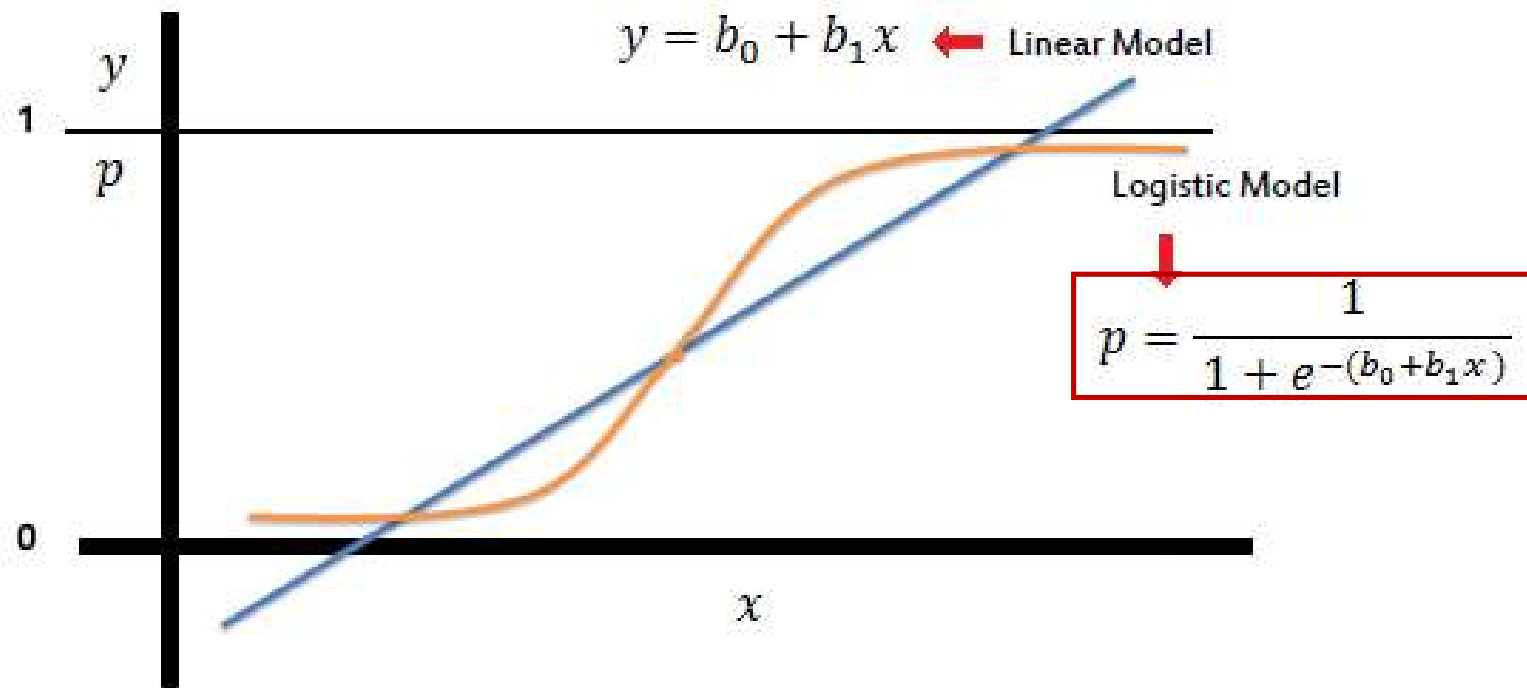
Issue of linear regression: $P(x)$ is a probability with values between 0 and 1 whereas $P_1x + P_0$ can take values outside $[0,1]$

The logistic function

- The logistic function is defined as : $logistic(x) = \frac{1}{1+e^{-x}}$
- $logistic(x)$ is always in $[0;1]$, so it can describe a probability
- Properties:
 - If $x \rightarrow +\infty, logistic(x) \rightarrow 1$ (class A)
 - If $x \rightarrow -\infty, logistic(x) \rightarrow 0$ (class B)
- Also called sigmoid or S-shape



Logistic vs linear regression



3

Hypothesis testing

3.1 – Statistical tests



Hypothesis testing

- Measure whether the data gives sufficient evidence to reject a hypothesis

Null hypothesis H_0 vs Alternative hypothesis H_A (H_1)

- H_1 = Hypothesis of interest
- Use a statistic that follows a certain distribution under H_0
 - *Name of the test = name of the statistics*
 - *Can we reject H_0 ? Not rejecting H_0 is different from proving it !*
- We calculate the statistics based on our data
- As we know the distribution, we can compute the CDF $p(X \leq x)$
- Decision based on a significance threshold α and the p-value = how likely the measurement comes from the null
 - *If $p < \alpha \rightarrow$ we reject H_0 and consider the test significant*

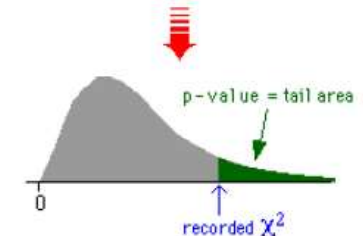
Summary statistic
(helps distinguish H_0 and H_A)

Test statistic
(standard distribution with no unknown parameters under H_0)

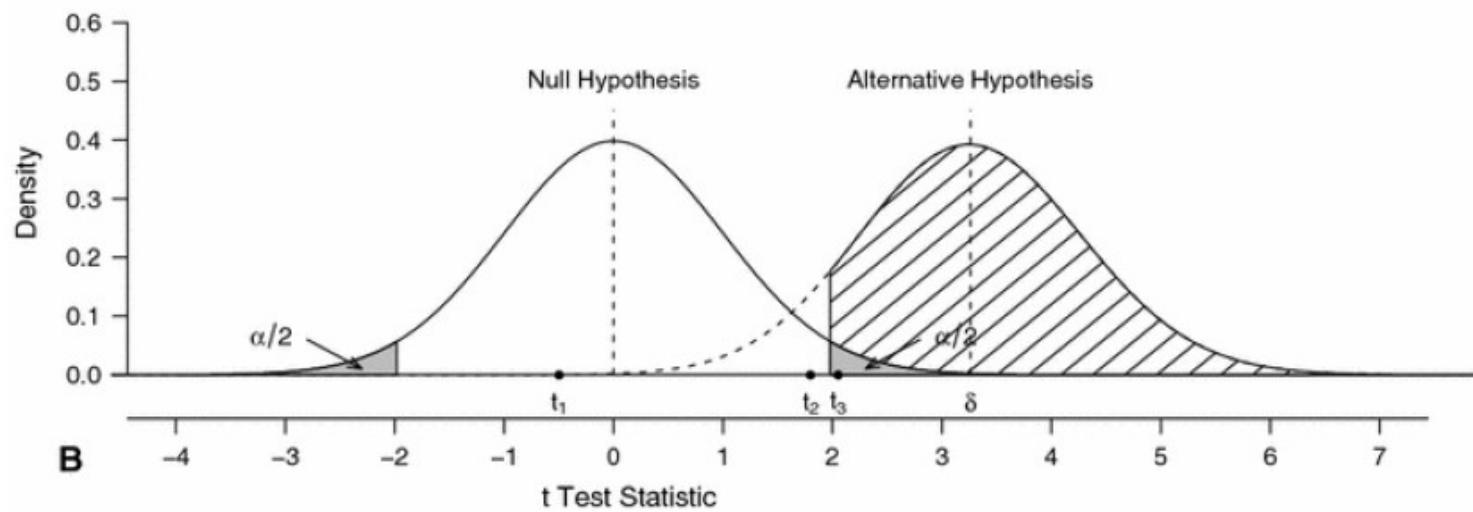
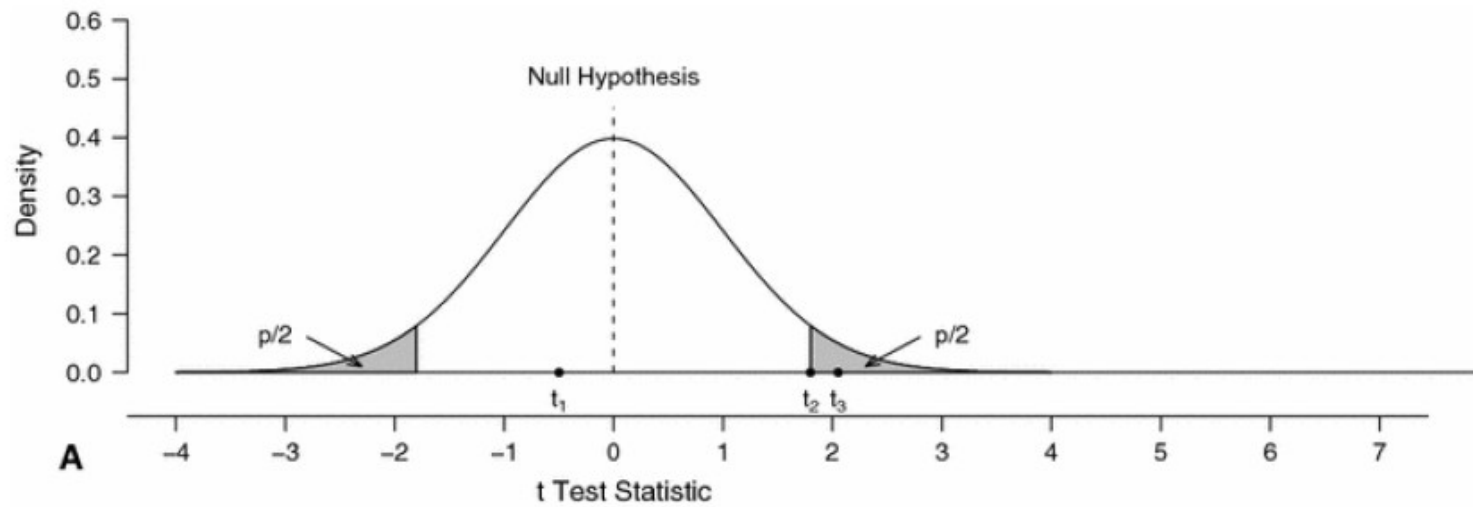
P-value
(probability of more 'extreme' test statistic)

$$\chi^2 = \sum \frac{(n_{xy} - e_{xy})^2}{e_{xy}}$$

$\chi^2 \sim$ chi-squared $((r-1)(c-1) \text{ df})$



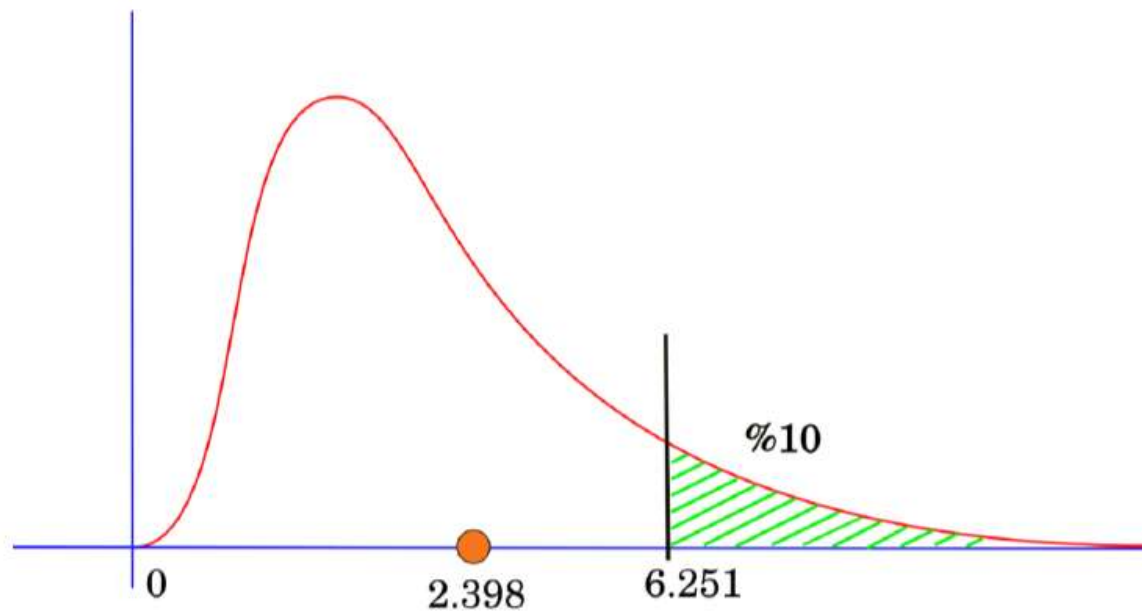
H_0 vs H_1



One-sided vs Two-sided test

- Depends on the hypothesis

<u>Two-sided:</u>	$H_0: " = "$	vs.	$H_1: " \neq "$
<u>One-sided:</u>	$H_0: " \leq "$	vs.	$H_1: " > "$
	$H_0: " \geq "$	vs.	$H_1: " < "$



3

Hypothesis testing

3.2 – Multiple testing

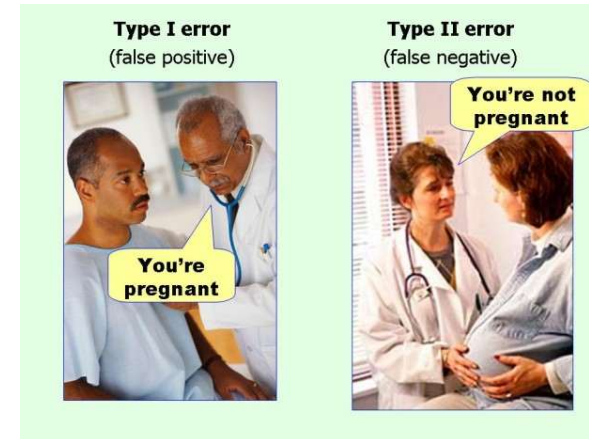
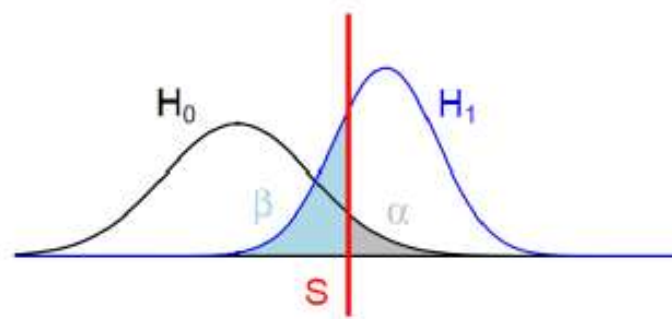


Multiple testing

- If $p < \alpha \rightarrow$ we reject H_0 and consider the test significant
- α corresponds to the type I error risk that we want to control

	H_0 is true	H_1 is true
H_0 not rejected	Correct decision	Type II error
H_0 rejected	Type I error = α	Correct decision

- If the number of tests increases, the risk increases
 $\rightarrow$ Need to take into account the multiple tests to maintain α at the desired level across all tests



Multiple testing: m tests

Family-Wise Error Rate (FWER)

- Bonferroni correction
- Simple to implement, harder to interpret

$$p_{critical} = \frac{0.05}{m}$$

- *“If all tests are under the null, probability that **one or more** of them is a false positive.”*

False-discovery based (FDR)

- Benjamini Hochberg procedure
- Harder to implement, easy to understand

$$p_{critical} = \operatorname{argmax}(p < \frac{i}{m} Q)$$

- $i = \text{rank}, Q = \text{FDR}$
- *“Proportion of significant tests that are false positives.”*

When to use which depends on

1. Best practices
2. Relative cost of a false negative/positive

Multiple testing: application in genomic studies

- Statistical significance:

- 5% for one test

- Genome-wide: one test per variant and per phenotype

$$\textit{phenotype} \sim \beta \times \textit{genotype} + \epsilon$$

- But all the variants are not independent and in reality we account for LD = correlation between the variants

- 5×10^{-8} for GWAS, 10^{-9} for sequencing based

$$p_{critical} = \frac{0.05}{m}$$

I➔ Exercise : Significance threshold

- If the adjusted genome wide significance threshold is 5×10^{-8} for GWAS, how many “effective” variants are there in a genotyped human genome?
- You are writing an article about a GWAS for 16 different traits. What will be your threshold for declaring significance?

$$p_{critical} = \frac{0.05}{m}$$

I➔ Exercise : Significance threshold

- If the adjusted genome wide significance threshold is 5×10^{-8} for GWAS, how many “effective” variants are there in a genotyped human genome? **10^6**
- You are writing an article about a GWAS for 16 different traits. What will be your threshold for declaring significance? **3.125×10^{-9}**

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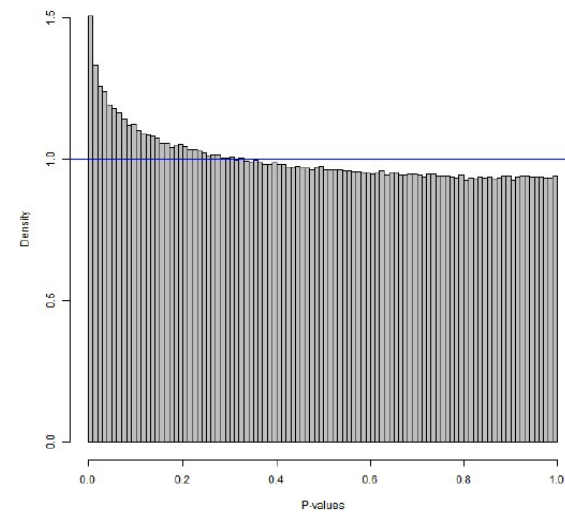
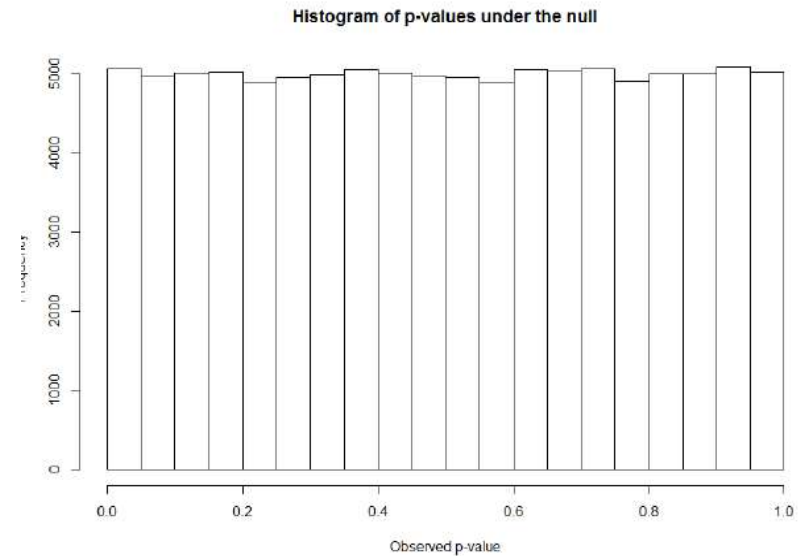
Hypothesis testing

3.3 – Checking the results

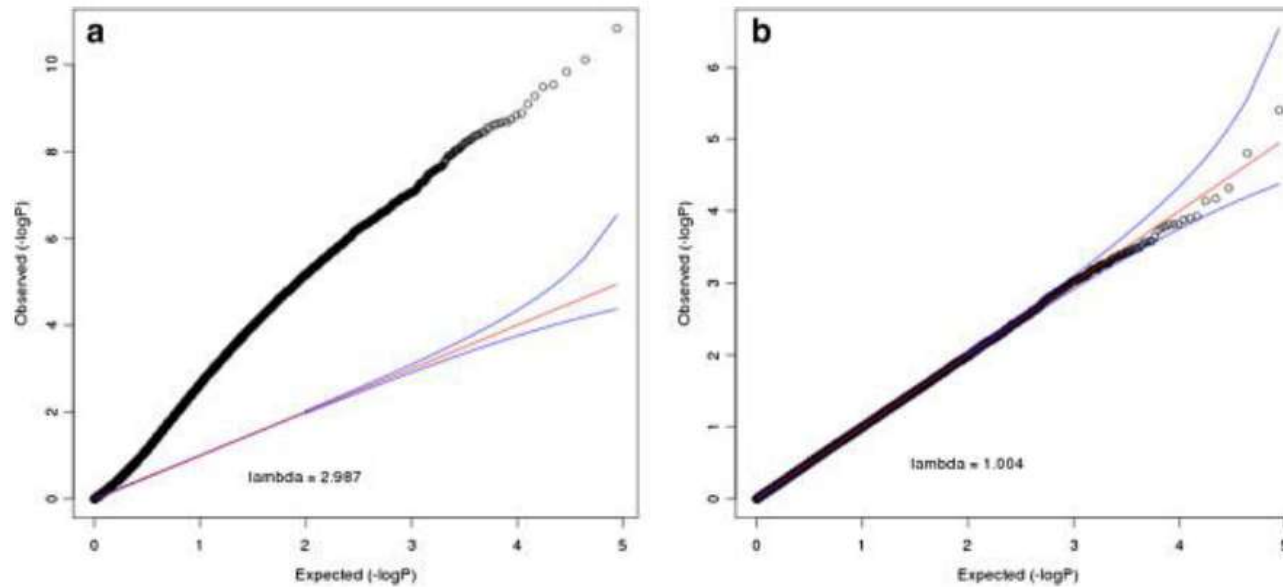


QQ-plots

- Under H_0 , p-values should be uniform [0,1]
- It is expected that most signals are around H_0
→ If we have much signal, more around 0
- Compare quantiles with expected ones: QQ-plot
- In R: *qqunif*



QQ-plots



- Inflation: too much signal
- Visual inspection but also lambda value

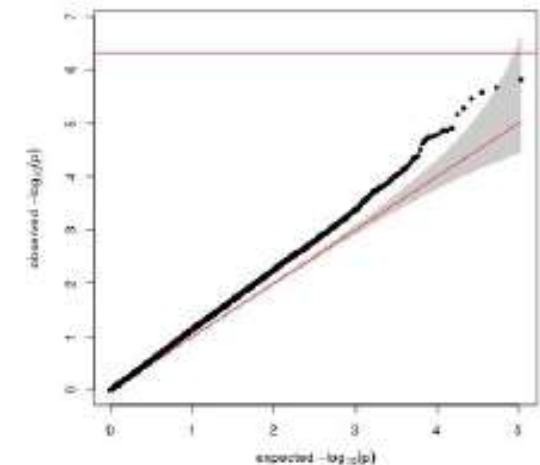
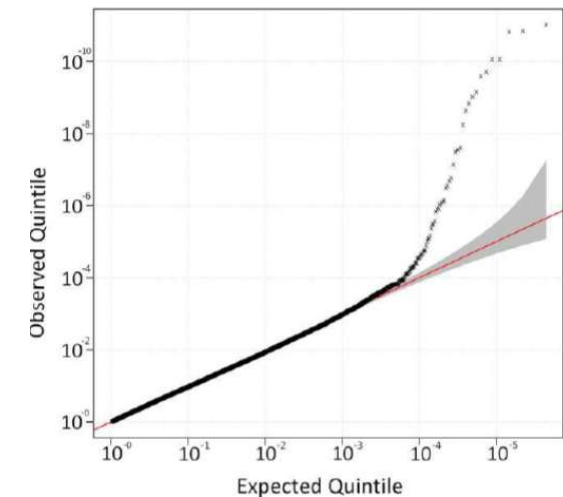
QQ-plots

- Appearance can be misleading:
 - A QQ-plot can seem inflated but just a lot of signal
 - And conversely...
- Calculation of the genomic inflation factor:

$$\lambda = \frac{\text{median}(Q_{\chi^2}(p))}{0.45}$$

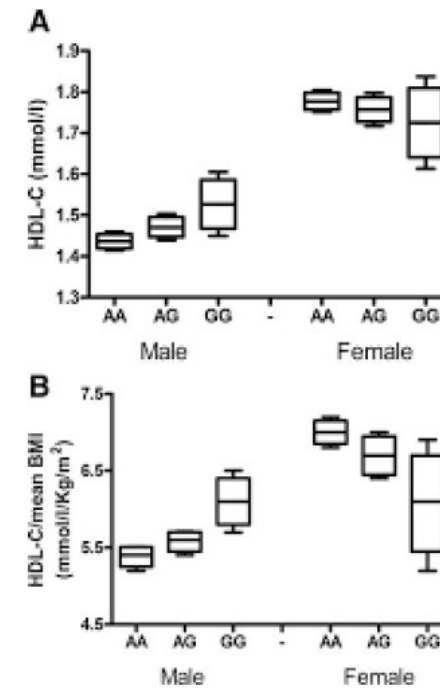
With 0.45 being the median of χ_1^2

- $\lambda > 1$: **inflation** (systematic bias)
- $\lambda < 1$: deflation (potential power issue)
- Ideally, we want to correct the model
- Can also adjust: GC correction: divide by lambda



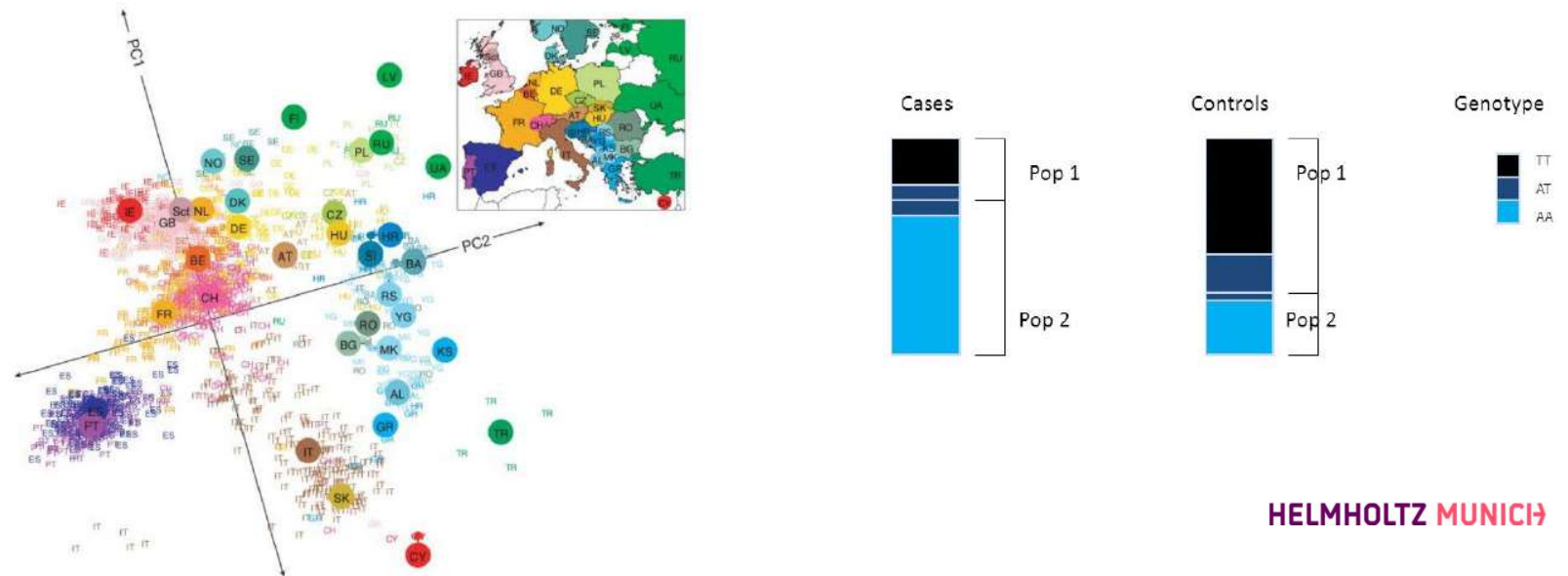
Problems in the model

- Covariates: sex, batch effects, chip effects
→ Potential bias if associated to phenotype and genotype



Problems in the model

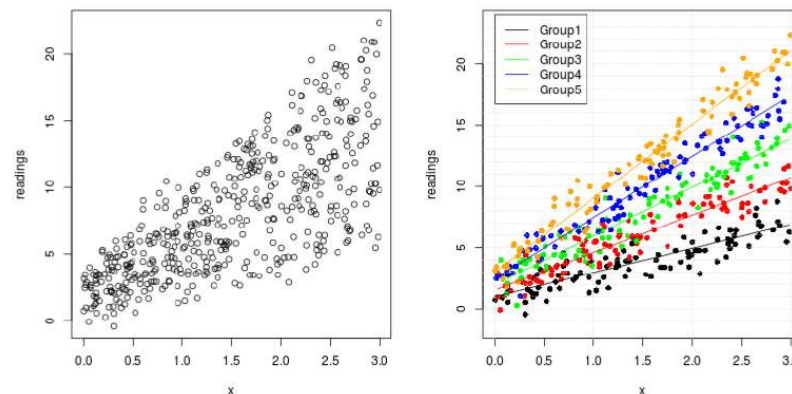
- Covariates: sex, batch effects, chip effects
 - Potential bias if associated to phenotype and genotype
- Structure or subpopulations
 - Allelic frequencies are known to be different from one population to another



Problems in the model

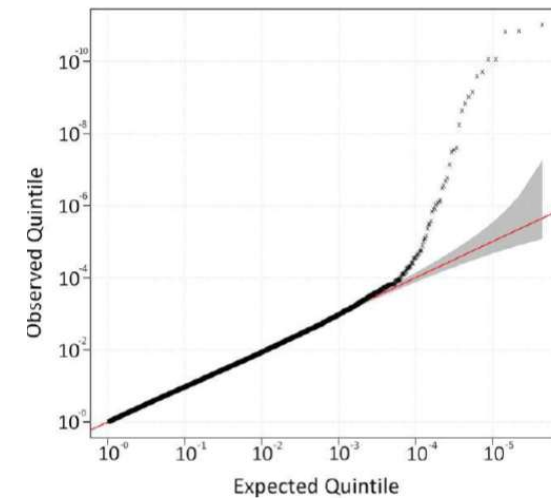
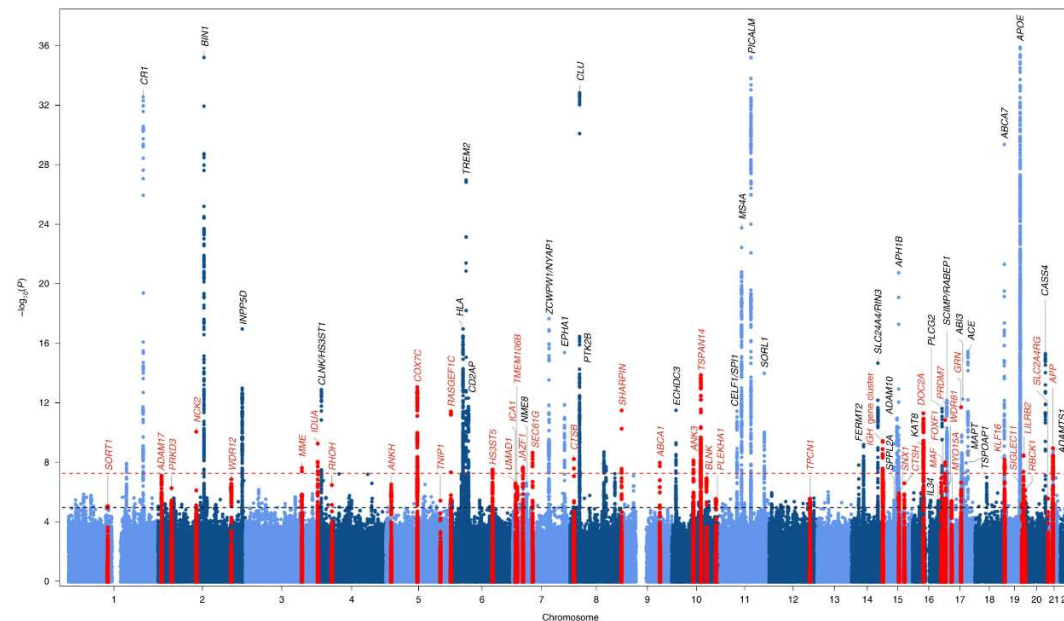
- Covariates: sex, batch effects, chip effects
 - Potential bias if associated to phenotype and genotype
- Structure or subpopulations
 - Allelic frequencies are known to be different from one population to another
 - Linear mixed models can model the intra-group effect
 - Adjust for principal components of a PCA

$$\text{phenotype} \sim \beta \times \text{genotype} + \beta_1 \times \text{covariates} + \beta_2 \times \text{structure} + \epsilon$$



Manhattan plot: visualization of the results

- Display $-\log_{10}(p)$ for every position in the genome
- Use a threshold (5×10^{-8}) to declare significance



4

Prediction



Prediction

- When we apply the estimated effects to new observations of the variable
- Suppose underlying assumptions
- Process = machine learning, predictive modelling, predictive analysis
- In human genetics, main task = **model effects of genotypes on phenotypes**
- Usually we do not predict
 - Except PRS (upcoming lecture)



Thank you.